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Department of Information technology
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[مُدِّخِر]

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1. Introduction

In today's fast-paced world, many individuals face challenges in managing their finances due to a lack of proper tools for tracking expenses and budgeting. This often leads to overspending and difficulties in achieving savings goals. This project aims to develop a user-friendly app that simplifies expense tracking, allows users to set personalized monthly budgets, and supports savings goals. With an intuitive design and smart reminders, the app will help users manage their finances effectively and stay motivated to reach their financial objectives.

1.1 Problem statement

Many people struggle with managing their daily expenses due to a lack of organization and tools to help track their spending and create personal budgets. This problem leads to exceeding the monthly budget and failing to achieve savings goals.

1.2 User Needs

1. An easy-to-use tool to track daily expenses.
2. A system to help create customized monthly budgets.
3. A way to clearly set and achieve savings goals.
4. An attractive visual design that motivates users to continue using the tool.

1.3 Project Goals

1. Design an easy-to-use and simple expense management app.
2. Enable users to log and track their expenses daily.
3. Provide a tool to create customized monthly budgets based on their needs.
4. Enhance the ability to achieve savings goals through smart alerts and reminders.
5. Deliver an enjoyable user experience through an attractive and practical design.

2. Methodology

2.1 Interviews

2.1.1 The required information

The interviews aimed to understand participants' challenges in managing personal expenses, their approaches to tracking spending, and their preferences for financial management tools. Key topics included budgeting difficulties, expense tracking habits, desired features in a financial management app, and preferences for notifications and financial progress tracking. These insights helped define user needs and guided the development of relevant app features.

2.1.2 The sample

The participants included a mix of students, employees, entrepreneurs, parents, retirees, and small business owners. Their selection was based on capturing diverse financial behaviors, from daily spending habits to long-term investment strategies. Each participant provided valuable input that helped refine our user personas and ensure the app's design aligns with different financial needs.

2.1.3 Selecting the sample

Participants were recruited through direct invitations and professional network referrals. The interviews were conducted through one-on-one sessions, either in-person or virtually, ensuring a comfortable setting for detailed discussions about financial habits and app expectations.

2.2 Questionnaires

2.2.1 The gathered information

The survey collected data on participants' financial habits, challenges in budget management, the desire for tools to categorize expenses, and their interest in receiving reminders or using visual tools to improve financial tracking and goal achievement.

2.2.2 The sample

A random sample was selected from users aged 18 and above, with diversity in social and educational backgrounds, to cover a wide range of needs and preferences. The sample was also designed to include individuals who currently use financial applications, as well as those who do not use them, in order to analyze the differences in their financial needs and preferences.

2.2.3 Reaching the sample

Participants were reached through an online survey distributed via Google Forms. The survey was shared on social media platforms.

3. Results

3.1 Interviews results

Interview Results:

Before designing our app, we conducted interviews with 8 individuals from diverse age and professional backgrounds who are looking to better manage their personal expenses and achieve their financial goals. Through a variety of questions, the interviews aimed to understand the users' needs and the challenges they face in managing personal expenses.

In the interviews, we found common issues such as the difficulty of tracking monthly expenses and the lack of a centralized tool for setting and monitoring budgets. There were also challenges in understanding the available funds across different spending categories. Our app, "Mudhakhir," helps address these issues by providing a flexible interface that allows users to log expenses, set budgets, and track various categories.

We also gathered feedback from users who provided their input on the app's design, asking them about their preferences for notification settings, such as alerts for exceeding the budget, and options for saving financial progress. They also expressed interest in adding a password protection feature for the app.

3.2 Questionnaires results



3.3 Discussion of the questionnaire results

1. Age Distribution

- Most participants fall between 25-44 years (50%), with 25% each in the younger (18-24) and older age groups (45+) categories.
- This suggests that the questionnaire is capturing data from a diverse but working-age audience, likely to be actively managing their finances.

2. Gender Distribution

- 55% of participants are male, and 45% are female, reflecting a fairly balanced distribution.
- This balance enhances the generalizability of financial behavior insights across genders.

3. Current Financial Goals

- The majority (35%) focus on saving, followed by equal interest in paying off debt and investing (25% each).
- This implies that a significant portion of respondents prioritize building a financial safety net while others aim to address debt or grow their wealth.

4. Difficulty in Sticking to a Monthly Budget

- Responses are evenly distributed, with 30% each finding it either “sometimes difficult” or “rarely difficult,” and only a small portion (10%) reporting it as “very difficult.”
- While most participants manage budgets moderately well, there’s room to help those who struggle with budget adherence.

5. Helpfulness of Expense Categorization Tools

- A large majority (55%) find automated categorization “very helpful,” while 30% consider it “somewhat helpful.”
- This highlights the importance of such tools in financial management, particularly in simplifying tracking and planning.

6. Awareness of Spending

- 55% of respondents know exactly where their money goes, while 35% have partial awareness.
- Although the majority maintain good spending awareness, the 35% who lack precise tracking could benefit from digital tools.

7. Effectiveness of Reminders for Tracking Expenses

- 45% always find reminders helpful, and 35% benefit from them occasionally.
- This implies that reminders play a crucial role in helping participants stick to their financial plans.

8. Impact of Having a Clear Savings Plan

- A majority (65%) feel motivated by having a clear savings plan, while 30% feel it helps them somewhat.
- This shows the motivational power of setting concrete savings goals.

9. Usefulness of Visual Tools for Managing Money

- 55% consider them very useful, and 30% find them somewhat useful.
- Visual tools like charts and graphs seem to play a significant role in simplifying complex financial information.

10. Effect of a Busy Schedule on Tracking Expenses

- 40% say their busy schedules impact their tracking ability, and 35% experience occasional disruptions.
- This result suggests that automation or streamlined tracking methods could address time-related barriers.

4. Personas

We have identified our personas based on interviews and questionnaire results.

We identified our first persona, “Norah”, as a 22-year-old Saudi university student from Jeddah. She struggles with tracking small daily expenses and managing a limited income. Norah relies on basic tools like notes or bank apps and prefers simple, automated budgeting help through push notifications and visual summaries. Her financial goals revolve around saving and staying within budget.

We identified Norah based on Interview 1 [A1, A4, A5, A6, A7, A10] + Interview 2 [A1, A5, A6, A7] and questionnaire results related to younger users (18–24), budget difficulties, and preference for visual tools and reminders.

The second persona is “Ahmed”, a 35-year-old Saudi entrepreneur from Riyadh, married with one child. He manages both personal and business finances and often struggles with balancing cash flow and budgeting. He relies on spreadsheets and seeks an automated system that offers syncing, alerts, and categorized spending insights. Security, real-time push notifications, and visual overviews are essential to him.

We identified Ahmed based on Interview 3 [A1, A2, A4, A5, A6, A10] + Interview 8 [A1, A4, A5, A6, A9, A10] and questionnaire results showing interest in savings, automation, and time-saving financial tools.

The third persona is “Fatimah”, a 60-year-old retired Saudi woman from Dammam. She manages her household finances on a fixed income and prefers simple, secure, and easy-to-use tools. Fatimah uses paper-based tracking but is open to digital tools that reduce manual effort. Her main concerns are medical expenses, savings longevity, and avoiding app complexity.

We identified Fatimah based on Interview 5 [A1, A4, A6, A8, A10] + Interview 7 [A1, A4, A5, A6, A9] and questionnaire responses from older participants (45+), particularly regarding reminders, savings plans, and simplicity in financial tools.

NORAH



PROFILE

Gender : Female
Age : 24
Location : Riyadh, Saudi Arabia
Marital Status : Single
Family : No children, living with family

GOALS

- Learn to manage personal expenses and start saving for long-term goals like travel or further education.
- Become financially independent as she starts her first full-time job after university.
- Track daily expenses more efficiently without spending too much time on manual budgeting.

NEEDS

- A simple, user-friendly app that automatically tracks spending and categorizes transactions.
- Tools to help set spending limits and receive alerts when nearing budget limits.
- Visual summaries like charts and graphs to make sense of spending patterns.

FRUSTRATIONS

- Difficulty tracking small, daily expenses (e.g., transport, food) which accumulate quickly.
- Trouble balancing fixed costs like bills and discretionary spending on personal items.
- Overwhelmed by unexpected expenses that make budgeting harder, such as outings with friends or textbooks for additional courses.

FATIMAH



PROFILE

Gender : Female
Age : 60
Location : Dammam, Saudi Arabia
Marital Status : Widowed
Family : Grandmother

GOALS

- Effectively budget her fixed retirement income to cover daily expenses and save for any medical or emergency costs.
- Maintain financial stability over the long term while ensuring her savings last.
- Simplify financial management with tools that require minimal effort.

NEEDS

- A simple, easy-to-use app with automatic expense categorization and reminders for when nearing budget limits.
- Clear financial reports that display essential expenses like medical, utilities, and groceries.
- Tools that help her set monthly savings goals and track them without needing extensive manual input.

FRUSTRATIONS

- Struggles to manage unexpected medical or household expenses on a fixed income.
- Difficulty keeping track of multiple recurring bills and auto-payments, leading to occasional overspending.
- Feels less comfortable with technology and prefers simple interfaces, avoiding complex apps with too many features.

AHMED



PROFILE

Gender : Male
Age : 35
Location : Jeddah, Saudi Arabia
Marital Status : Married
Family : Father of two children

GOALS

- Effectively manage both personal and business finances to ensure business growth while maintaining family financial stability.
- Improve cash flow management during slow business months and save for long-term personal goals like children's education.
- Automate expense tracking and minimize manual work on financial management.

NEEDS

- An app that tracks both business and personal expenses, with clear reports and categorizations.
- Real-time alerts when approaching budget limits and tools to set financial goals for the business and family.
- Security features to protect sensitive business data and ensure no misuse of personal or business funds.

FRUSTRATIONS

- Difficulty separating business expenses from personal ones, leading to cash flow issues.
- Overwhelmed by managing personal finances while growing a business, especially with unexpected business expenses.
- Reliance on manual spreadsheets and inconsistent tracking methods, which take up valuable time.

5. Solution

Our proposed solution is a smart expense tracking app that simplifies budgeting, enhances financial awareness, and supports users in achieving their financial goals.

5.1 Key Features & Functionalities

1. Manual Expense Tracking & Categorization

- User-Entered Transactions: Users manually input their expenses to maintain full control over their data.
- Customizable Categories: Users can categorize expenses (e.g., food, utilities, entertainment) based on their preferences.
- Quick Entry Feature: A simplified interface allows for fast and easy expense recording.

2. Smart Budgeting & Spending Limits

- Set Monthly Budgets: Users can allocate a budget for different spending categories.
- Spending Alerts: Notifications when spending reaches 80%, 90%, and 100% of the budget.
- Insights & Recommendations: The app provides personalized suggestions to reduce overspending

3. Financial Goals & Savings Management

- Goal-Setting Feature: Users can define savings goals (e.g., vacation, emergency fund).
- Progress Tracking: Visual indicators show how close users are to reaching their goals.
- Automatic Round-Up Savings: The app can round up transactions and save the difference.

4. Visual Financial Reports & Insights

- Charts & Graphs: Users get easy-to-understand visual breakdowns of spending patterns.
- Weekly & Monthly Summaries: A snapshot of income vs. expenses, highlighting areas for improvement.
- Expense Trends Analysis: The app identifies patterns in spending habits to help users adjust.

5. Smart Notifications & Reminders

- Push Notifications: Alerts about bill due dates, low balances, and unusual transactions.
- Daily/Weekly Spending Summary: Users receive a short recap of their spending trends.
- Customizable Notifications: Users can choose notification frequency and type.

6. Strong Security & Privacy Measures

- End-to-End Encryption: Ensures that user data is protected from breaches.
- Two-Factor Authentication (2FA): Extra security layer for login and sensitive actions.
- Minimal Data Storage: No long-term storage of sensitive financial information.

7. User-Friendly Interface & Experience

- Simple & Intuitive Design: A clean, easy-to-navigate UI with minimal learning curve.
- Ad-Free Experience: No intrusive ads to maintain a smooth user experience

6. User stories

1- As a user, I want to create and name my own savings goals and assign money to them so that I can track my progress easily.

2- As a user, I want the app to analyze my income and expenses and suggest a smart budget so that I can manage my money better.

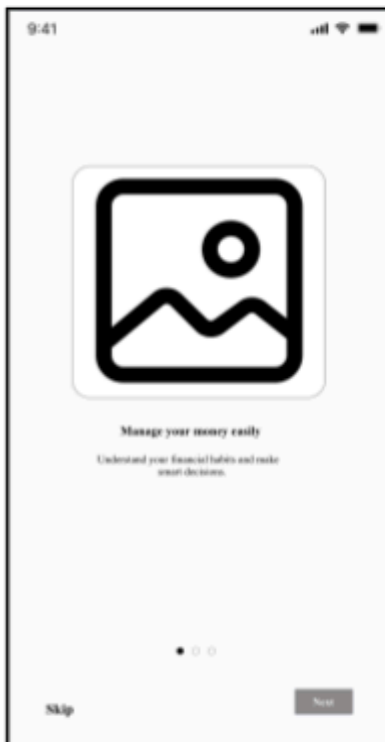
3- As a user, I want an emergency fund tracker that helps me set aside money for unexpected expenses so that I am financially secure.

4- As a user, I want to track my income and expenses in different categories so that I can understand where my money is going.

7. Sketches

Onboarding & Account Access

Sketch 1



Fighter 7.1

Sketch 2



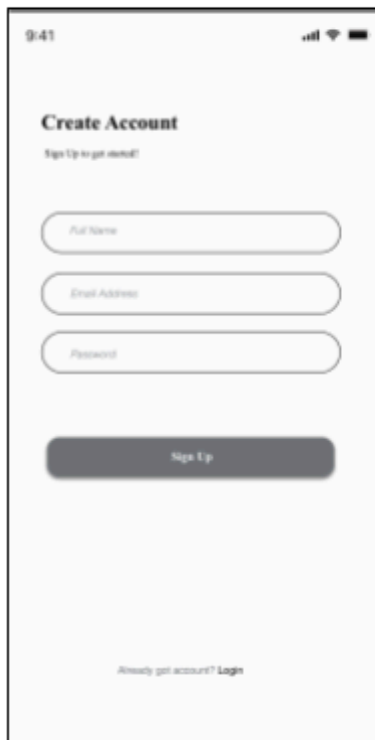
Fighter 7.2

Sketch 3



Fighter 7.3

Sketch 4



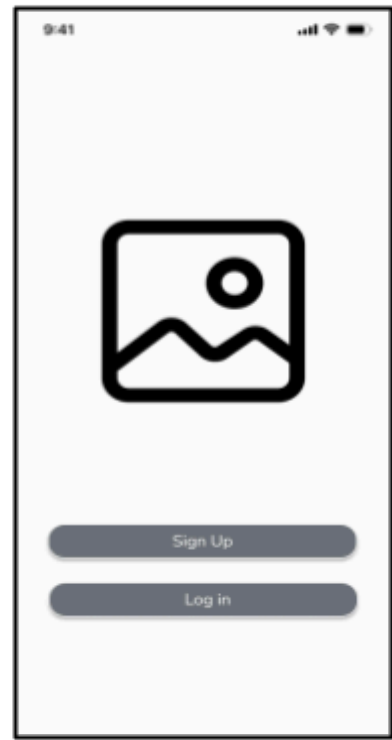
Fighter 7.4

Sketch 5



Fighter 7.5

Sketch 6



Fighter 7.6

Forgot Password

Sketch 7



9:41

⌕



Please enter your email address to receive a verification code

Email

Next

Fighter 7.7

Sketch 8



9:41

⌕



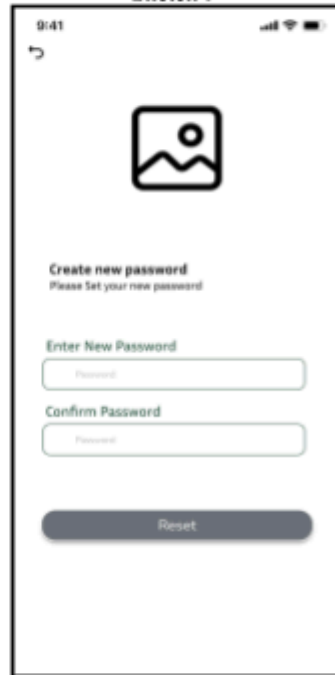
Check Your Inbox

A 4-digit confirmation has been sent to your email

Next

Fighter 7.8

Sketch 9



9:41

⌕



Create new password

Please Set your new password

Enter New Password

Confirm Password

Reset

Fighter 7.9

Sketch 10



9:41

⌕



Password Updated!

Your password has been changed successfully

Log in

Fighter 7.10

Settings



Fighter 7.11



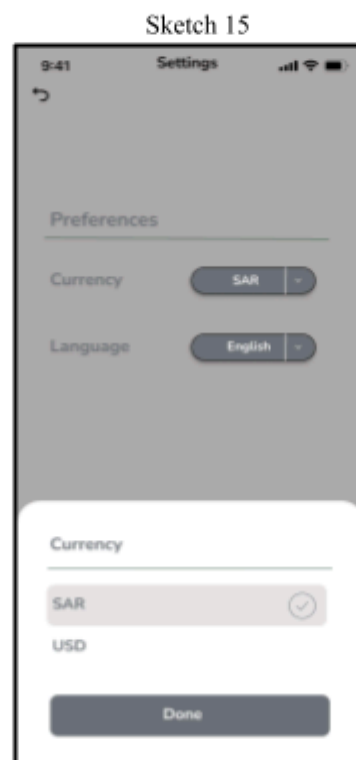
Fighter 7.12



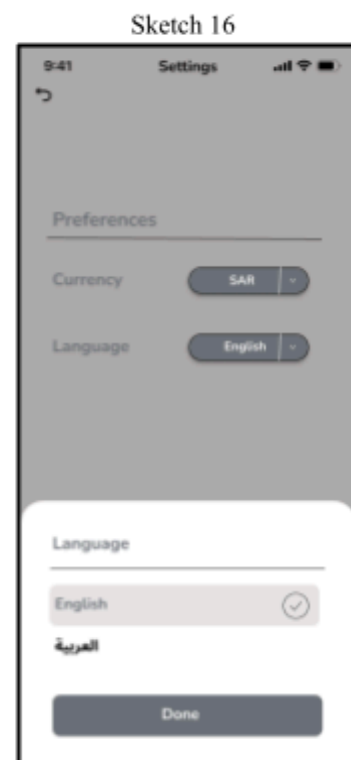
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Fighter 7.14

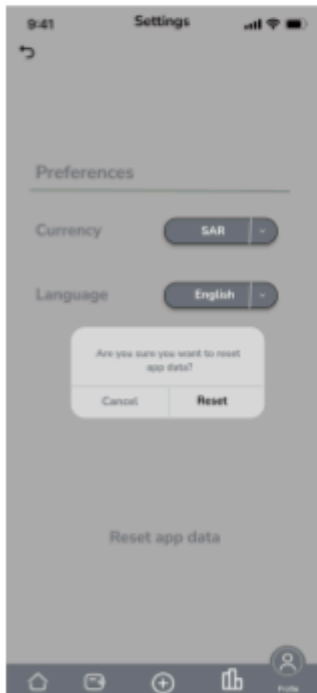


Fighter 7.15



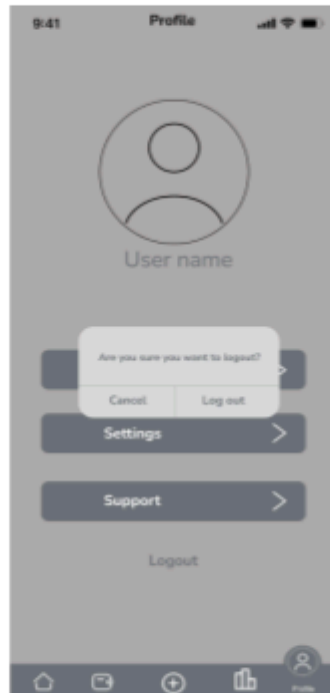
Fighter 7.16

Sketch 17



Fighter 7.17

Sketch 18

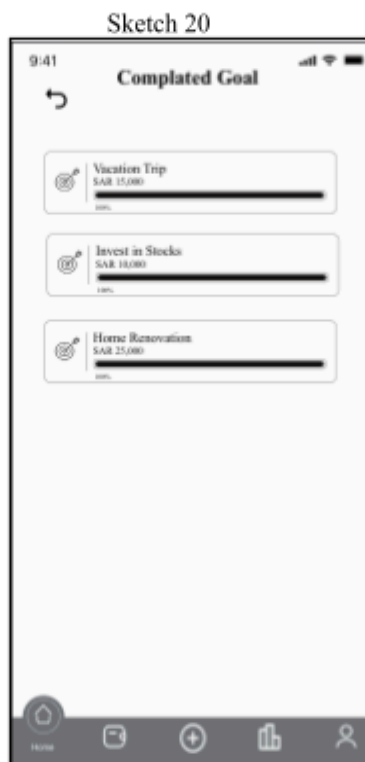


Fighter 7.18

Dashboard & goal management



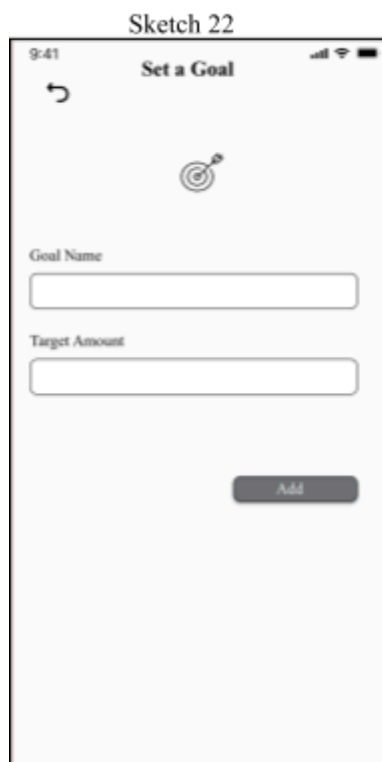
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Fighter 7.20



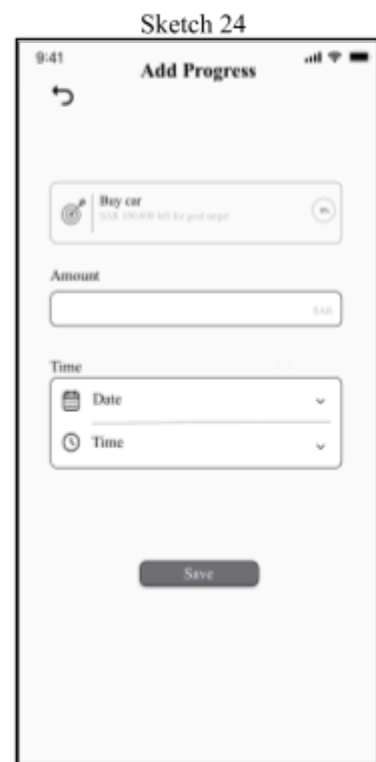
Fighter 7.21



Fighter 7.22



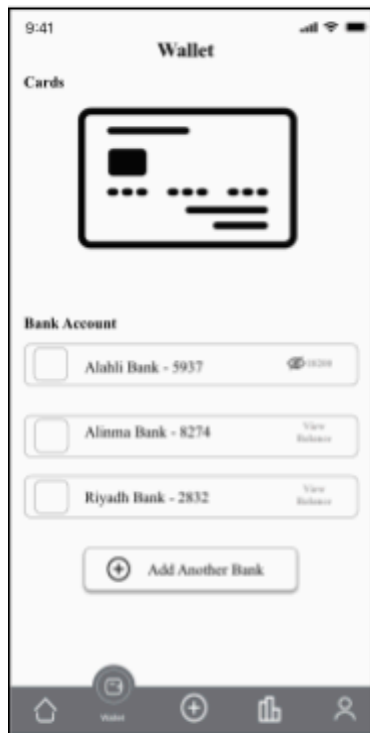
Fighter 7.23



Fighter 7.24

Accounts & Wallet Management

Sketch 25



Fighter 7.25

Sketch 26



Fighter 7.26

Sketch 27



Fighter 7.27

Sketch 28

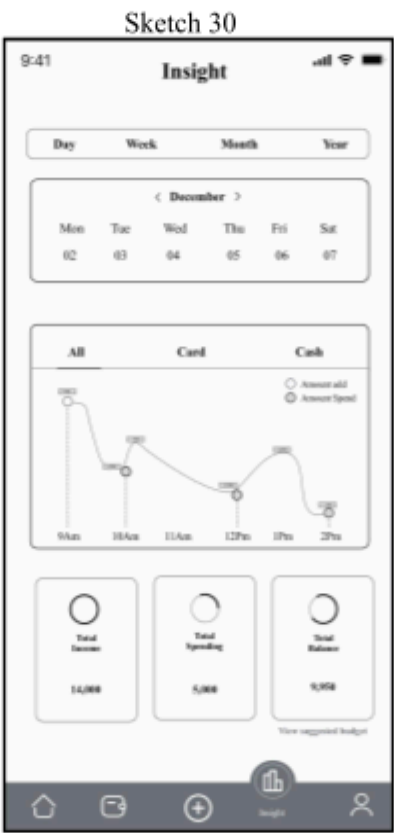


Fighter 7.28

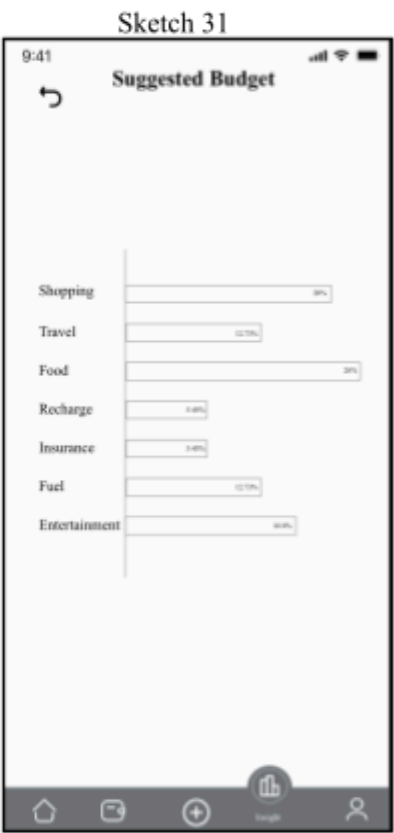
Budget & Expense Management



Fighter 7.29



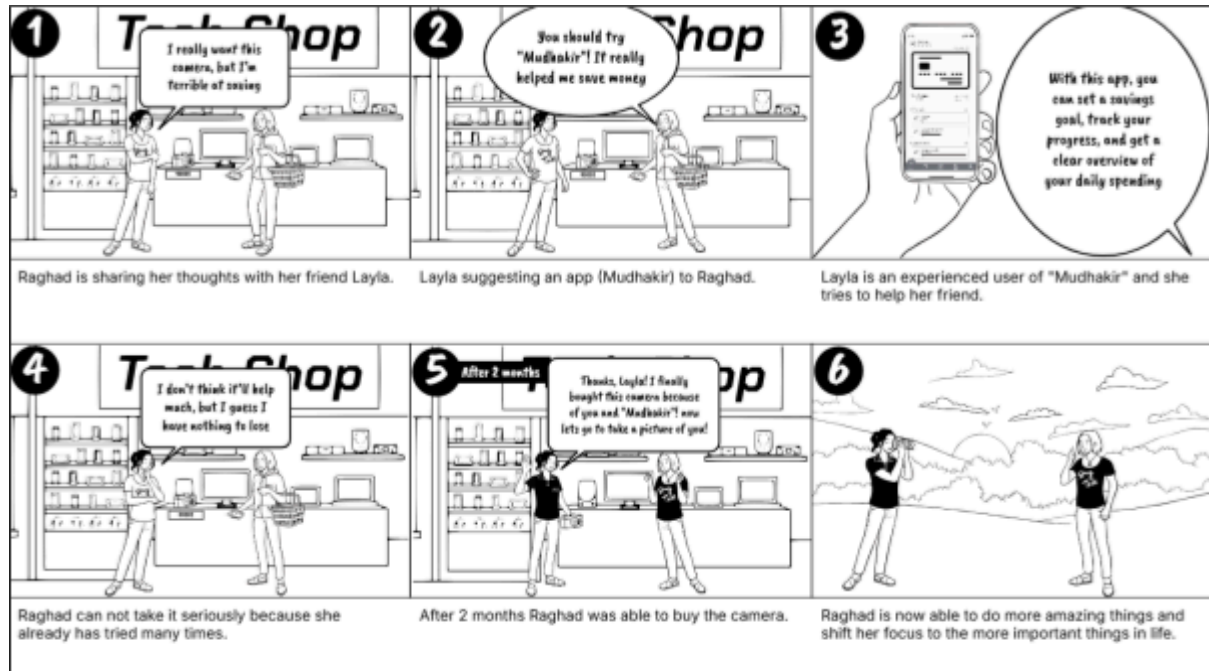
Fighter 7.30



Fighter 7.31

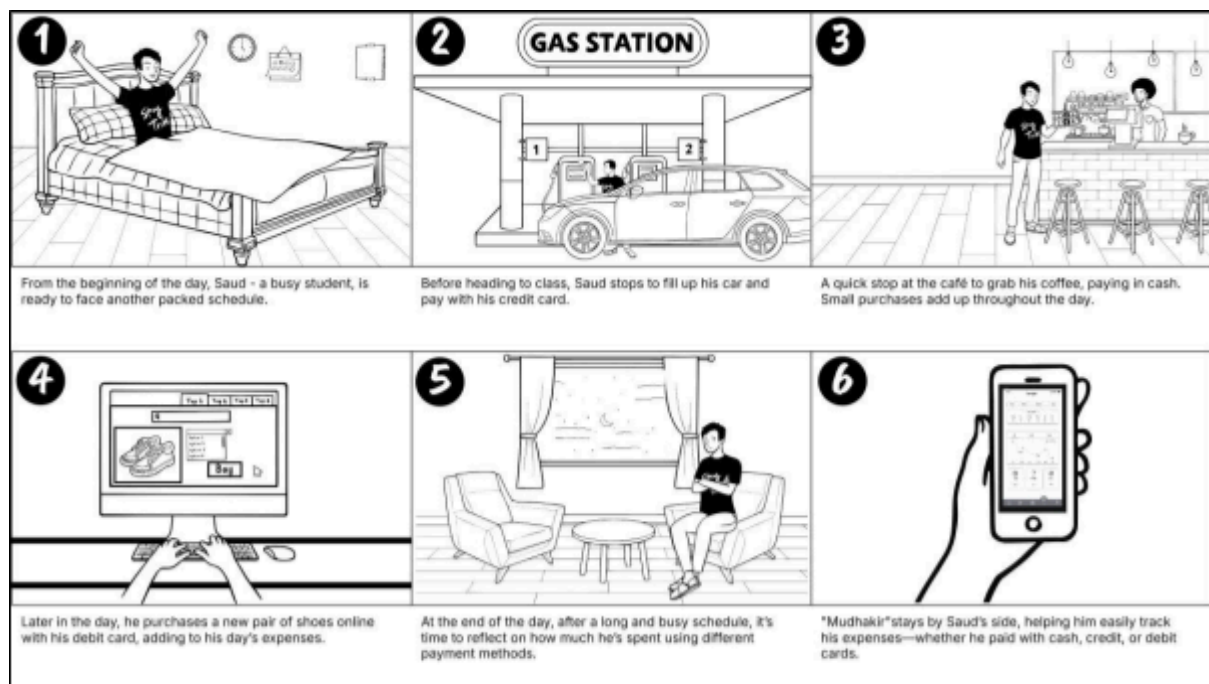
8. Story Board

8.1 Story Board



Fighter 8.1

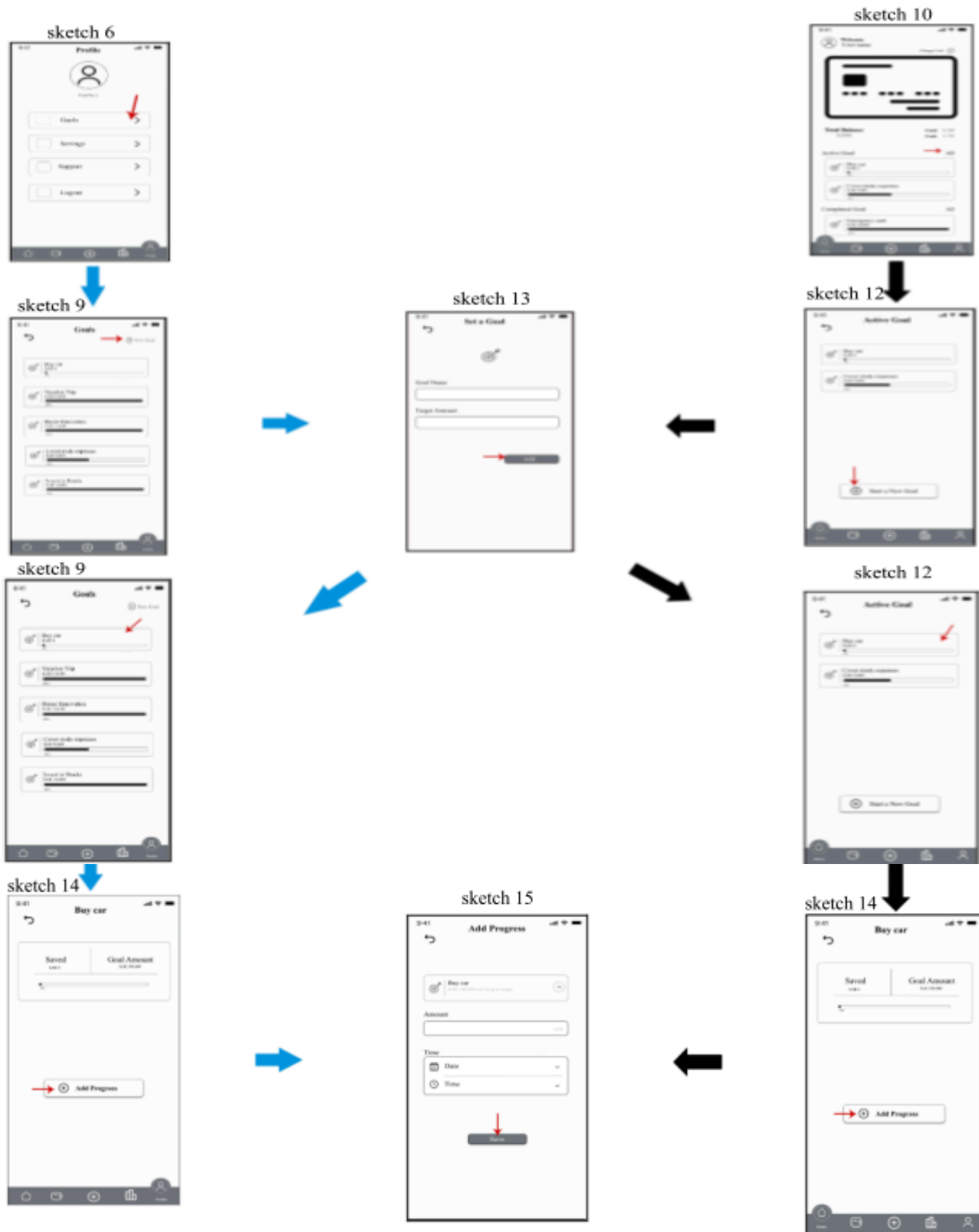
8.2 Story Board



Fighter 8.2

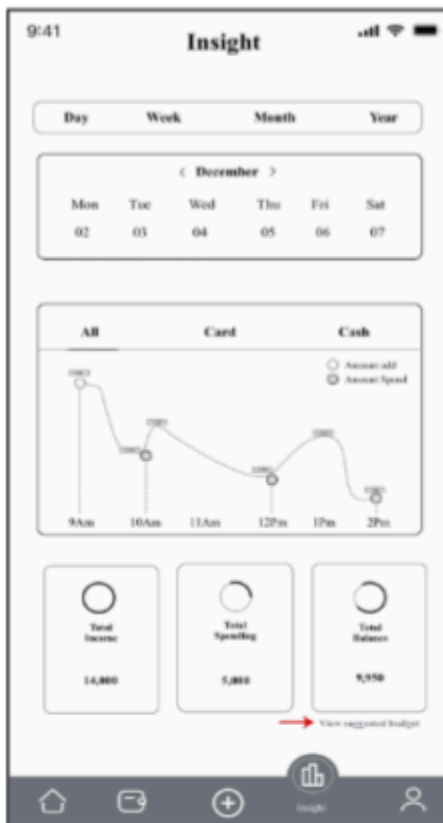
9. User Flow

9.1 User Flow - As a user, I want to create and name my own savings goals and assign money to them so that I can track my progress easily.



9.2 User Flow - As a user, I want the app to analyze my income and expenses and suggest a smart budget so that I can manage my money better.

sketch 21



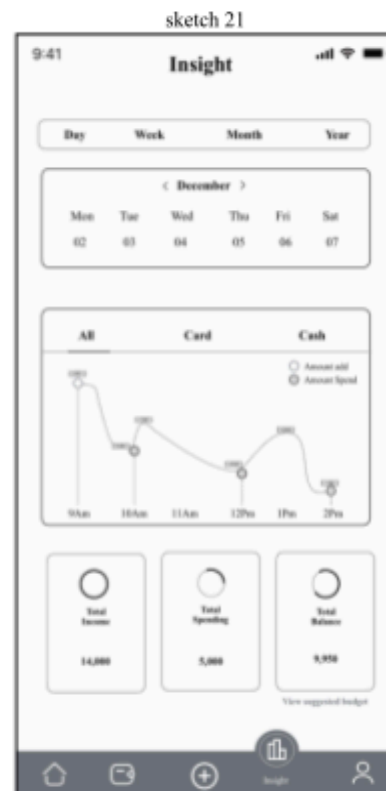
sketch 22



9.3 User Flow - As a user, I want an emergency fund tracker that helps me set aside money for unexpected expenses so that I am financially secure.



9.4 User Flow - As a user, I want to track my income and expenses in different categories so that I can understand where my money is going.



10. Design Rules

10.1 Learnability Principles

Rule	Principle		Sketch No/ User Flow No	Describe how you applied this principle
1 2	Predictability		Sketch 4	where the user creates an account, the flow of actions is predictable as the user is presented with a simple form to fill out, leading them to the next step after they register.
		Operation visibility	Sketch 14	demonstrates this well with a clear display of multiple savings goals, which users can select and track progress. The visible progress bars help make the operations of tracking goals clear.
3	Synthesizability		Sketch 28	shows adding funds to accounts, where users can synthesize information from their bank account details and input amounts to add funds.
4	Familiarity		Sketch 11	The Profile screen looks like a typical settings page found in many apps, making it easy for users to navigate and manage their preferences.
5	Generalizability		Sketch 30,31	showcase budgeting and expense management features, which could be generalized across different financial tracking tasks, making the app versatile.
6	Consistency		Sketche 25, 26, and 27	The navigation bar follows a consistent format with similar elements, which helps users understand how to move through the app, regardless of the screen they're on.

10.2 Flexibility Principles

[illegible]

10.3 Robustness Principles

Rule	Principle		Sketch No/ User Flow No	Describe how you applied this principle
12 13 14 15	Observability	Browsability	Sketch 26	The “Account” screen allows users to easily browse and observe their account details and transaction history. It provides clear visibility of balances and allows users to review the information quickly.
		Defaults	Sketch 27	When adding a new account, default fields are pre-set, like the “Bank Name” and “Card Number.” Users can quickly add a new card without needing to fill every detail manually.
		Reachability	Sketch 14	In the “Goals” section, users can access and manage their financial goals from the main screen with just a couple of taps. The section is readily reachable through the app’s navigation menu.
		Persistence	Sketch 29	The “Add Cash Expense” screen shows persistent information like the total cash balance, ensuring that users can easily continue entering expenses without losing track of their balance.
16	Recoverability		Sketch 31	In the “Suggested Budget” section, if a user accidentally loses their input or there is an error, they can always reset and recover the default suggested budgets and categories.
17	Responsiveness		Sketch 23	In the “Buy Car” section, the user can see live progress updates as they add funds, showing responsive and real-time feedback.
18	Task Conformance		Sketch 19	The “Welcome” and dashboard section shows a breakdown of active goals, ensuring that each goal aligns with the tasks, making it easy for users to track the right tasks.

11. Prototype

11.1 Features

11.1.1 User Authentication & Account Management

The application allows users to sign up, log in, and reset their passwords through a secure authentication system. It also includes an email-based password recovery mechanism to ensure easy account access.

11.1.2 Wallet & Bank Account Management

Users can link their bank accounts and cards, view balances, and add new accounts. They can also track their savings progress with ease.

11.1.3 Goal Setting & Financial Planning

Users can set financial goals, such as saving for a car or a vacation, with visual progress indicators and the ability to allocate funds toward these goals over time.

11.1.4 Expense Tracking & Budgeting

The app features a detailed expense tracking system, enabling users to log daily expenses and categorize them into groups such as shopping, travel, food, entertainment, insurance, fuel, and recharges. Additionally, it provides smart budget suggestions based on past spending habits.

11.1.5 Financial Insights & Reports

The application generates real-time financial insights, allowing users to analyze spending trends over different periods (daily, weekly, monthly, yearly). Users can also view interactive charts and summary reports to help them understand their financial habits and make informed decisions.

11.1.6 Emergency Fund & Savings Management

Users can create a dedicated emergency fund to track savings separately from their other financial goals. The app provides insights into how much they should save based on their income and spending patterns.

11.1.7 User Profile & Settings

The application allows users to customize their profile settings, change language and currency preferences. Additionally, users can access customer support when needed.

11.2 Tools

11.2.1 Figma

We used Figma to design the prototype. Figma is an online interface design tool that offers powerful features for prototyping and encourages seamless collaboration.

11.2.1 Removebg

We utilized remove.bg, an online tool designed for automated background removal from images. This tool leverages AI-powered processing to efficiently detect and remove backgrounds while maintaining the quality of the subject.

11.3 YouTube Video

<https://youtu.be/YITi97jAhFI>

11.4 Prototype Design

<https://www.figma.com/proto/rKY41QOpTrv5C9PN6oNDEK/Mudhakhir-edu?node-id=4-297&t=qZpcwqUPffdGlccC-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=4%3A297>

12. Usability Testing

12.1 Participants

We have 6 participants in our study, their ages ranging between 19 and 27. Most of them have not used any similar apps/ websites. However, all of them have at least the basic knowledge to operate a device and navigate through a mobile app or a website interface.

The study sample contains people from different backgrounds, 2 males and 4 females:

Mohammed is a recent-graduate computer science, Wojoud is a high school math teacher, Shahad is a housewife, Sara is a marketing professional, Saad is a last-year university student, and Huda is a university freshman.

<i>Participant name</i>	<i>Age</i>	<i>Gender</i>	<i>Experience with similar products (1-5)</i>	<i>Use of mobile/website interface (1-5)</i>
Mohammed	23	M	3	5
Wojoud	27	F	3	5
Shahad	25	F	1	5
Sara	24	F	2	5
Saad	22	M	2	5
Huda	19	F	1	5

12.2 Procedure

12.2.1 Outline of the procedure

After we completed our application “Mudhakhir” prototype, we have carefully chosen the 5 most important tasks that cover the main functions of the application.

To measure the ease and simplicity of accomplishing the tasks, we have wrote a satisfaction questionnaire that contains 5 closed questions to measure the ease and simplicity of application, then we planned and wrote the facilitator script which is written carefully to read it infront of the participants later, then we also have written a consent form to take the participants agrrement.

We have chosen 6 random people to conduct the usability test. At the beginning of each session, we welcomed the participants and took the chance to get to know them better. After that, we asked them to fill out the consent form.

Once the consent is signed, we handed the participants the prototype of our application and read the facilitator script aloud. During the session, we asked them to use the Think Aloud Technique, encouraging them to talk and express what they are doing and what they believe is happening while completing the given 5 tasks.

After completing the tasks, each participant was asked to fill out the satisfaction questionnaire to asses the overall experience.

12.2.2 Timeline

We conducted the usability test between April 6 and April 11, 2025, and the session times varied between 7 AM and 5 PM to meet the participants’ availability and preferences. The duration of each session was in the range of 10 minutes to 25 minutes.

<i>Participant name</i>	<i>Day</i>	<i>Date</i>	<i>Start session time</i>	<i>Finish session time</i>	<i>Duration of session</i>
Mohammed	Sun	6/4/2025	1:00 PM	1:10 PM	10 min
Wojoud	Mon	7/4/2025	7:32 AM	7:47 AM	15 min
Shahad	Mon	7/4/2025	9:26 AM	9:41 AM	15 min
Sara	Tue	8/4/2025	3:00 PM	3:12 PM	12 min
Saad	Wed	9/4/2025	12:00 PM	12:15 PM	15 min
Huda	Fri	11/4/2025	4:30 PM	4:55 PM	25 min

12.2.3 Tasks

5 tasks were given for the participants to complete using our “Mudhakhir” application to assess the usability of the application interface by measuring the navigation, structure, and readability.

<i>Task</i>	<i>Task</i>	<i>Objective of the task</i>
1	Add a bank account (Alahli Saving Account).	This task aims to measure the ability to add a new bank account.
2	Add an expense (for a 1000 SAR digital camera).	This task aims to configure how to add an expense.
3	View the Suggested Budget.	This task aims to measure the capability to locate and interpret the Suggested Budget.
4	Set a new goal (buying a laptop with 8000 SAR).	This task aims to measure the ability to set a new financial goal.
5	View the amounts spent and added to the cards.	This task aims to measure the capability to view the amounts spent and added to the card specifically.

12.2.4 Performance Measures and Feedback

To evaluate the usability of our application, we focused on three core performance metrics: effectiveness, efficiency, and satisfaction. Effectiveness was measured by observing how many of the assigned tasks each participant was able to complete successfully, and we also noted any errors made during the process. These errors were analyzed to identify potential issues in the interface or user flow. Efficiency was determined by measuring the amount of time it took each participant to complete the tasks, which gave us a clear picture of how intuitive and streamlined the interactions were. For satisfaction, we distributed a short feedback form after each task, allowing users to rate their experience based on clarity, ease of use, and overall comfort while interacting with the app.

Metric	How was it measured?	How was the data created/captured?
Effectiveness	Task success	By observing whether participants were able to successfully complete core tasks such as adding expenses, viewing suggested budgets, and tracking transactions within the app.
	Number of Errors	By recording user mistakes or confusion during testing sessions, such as incorrect entries or navigation issues.
Efficiency	Time on Task	Recorded the time each participant took to complete tasks within the app and calculated the average completion time.
Satisfaction	User surveys	Participants completed a survey after testing, rating their experience on ease of use, design clarity, usefulness, and overall satisfaction with the app.

12.3 Results

12.3.1 Task Completion Success Rate and Time on Task

All participants successfully completed all the tasks: Task1(Add a bank account), Task2(Add an expense), Task3(View the Suggested Budget), Task4(Set a new goal), and Task5(View the amounts spent and added to the cards).

Task 4 required participants to navigate multiple steps to set a new goal, as a result, it took the longest time to achieve (avg = 53.5 seconds). However, completion times for all completed tasks ranged from 26.5 to 53.5

Participant	Task 1		Task 2		Task 3		Task 4		Task 5	
	complete task	Time on Task	complete task	Time on Task	complete task	Time on Task	complete task	Time on Task	complete task	Time on Task
P1	✓	45	✓	52	✓	35	✓	31	✓	22
P2	✓	42	✓	32	✓	24	✓	39	✓	21
P3	✓	72	✓	48	✓	36	✓	60	✓	24
P4	✓	40	✓	55	✓	30	✓	70	✓	35
P5	✓	47	✓	38	✓	29	✓	57	✓	26
P6	✓	51	✓	45	✓	31	✓	64	✓	31
Completion Rates	6/6 100%		6/6 100%		6/6 100%		6/6 100%		6/6 100%	
AVG . Time on Task		49.5		45		30.8333		53.5		26.5

12.3.2 Errors

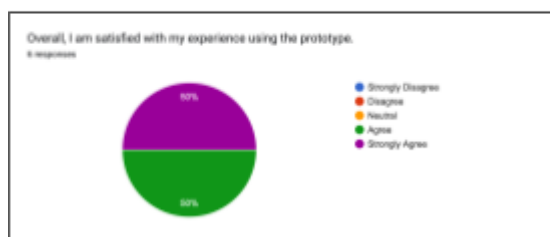
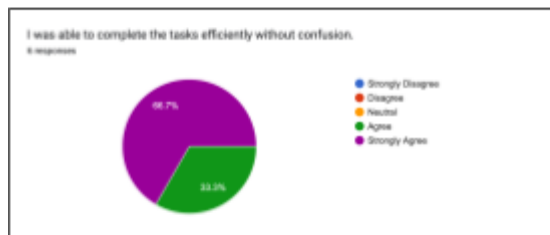
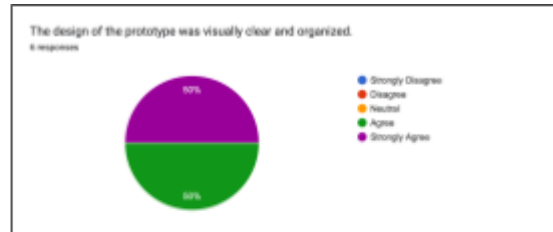
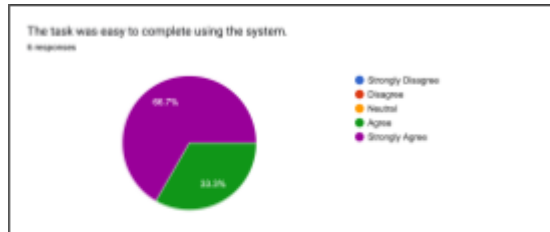
Tasks: 1,2,4, and 5 have been done seccussfly. Hoever task 3 has 3 participants who did a non-critical error.

Tasks	Number of errors	Errors description	(C,NC)
Task 1	0		
Task 2	0		
Task 3	3	The user overlooked the "View Suggested Budget" button due to its small and faded text.	NC
Task 4	0		
Task 5	0		

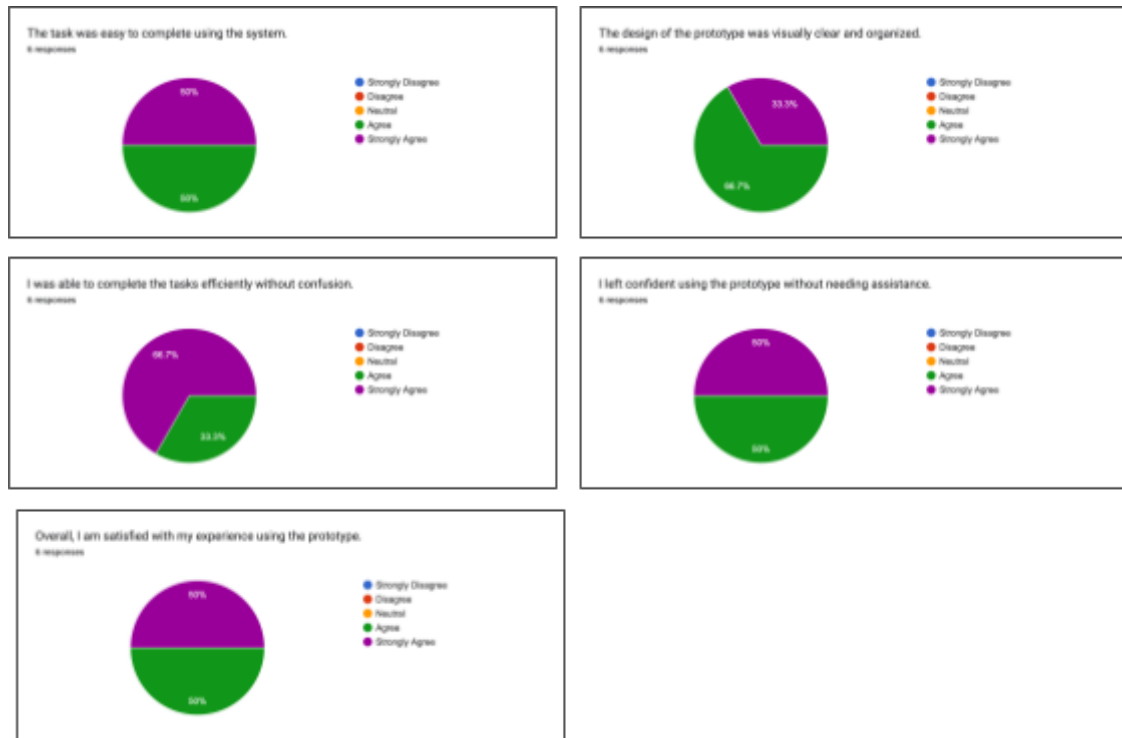
12.3.3 Satisfaction questionnaire

Graphs of results:

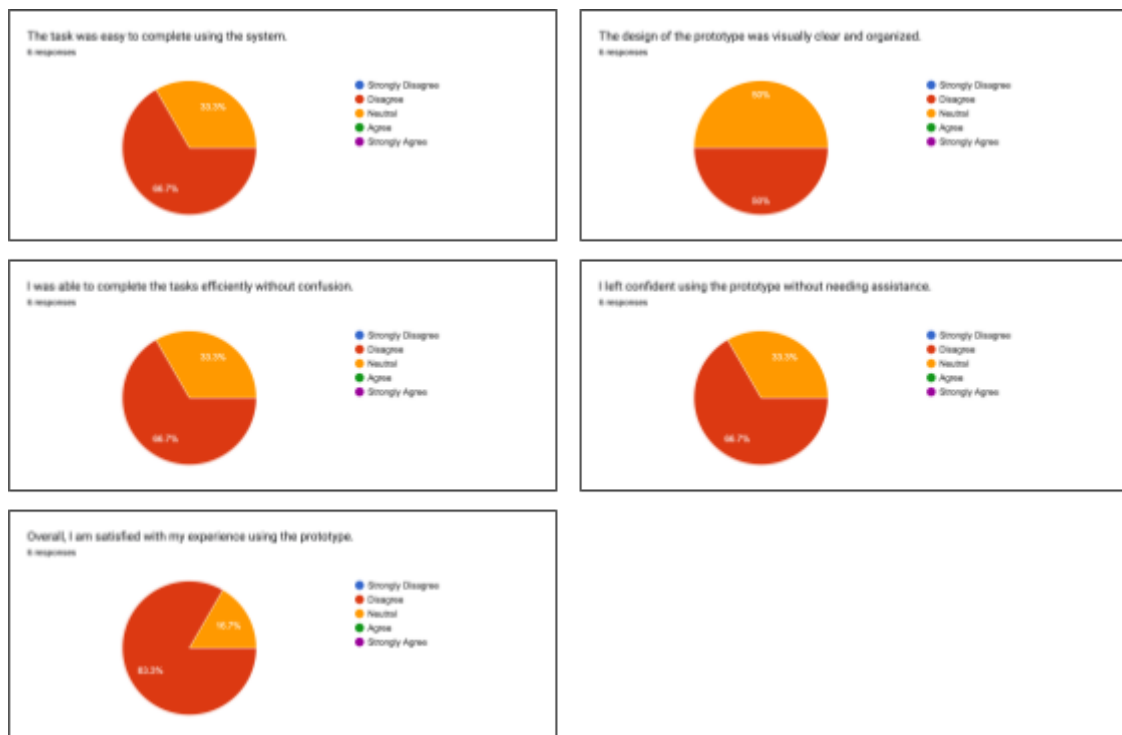
Task 1



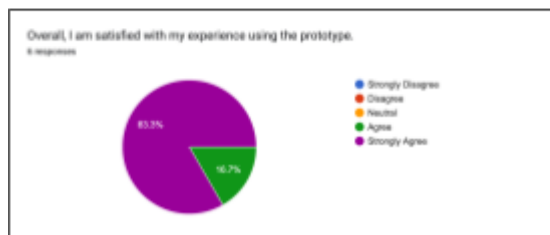
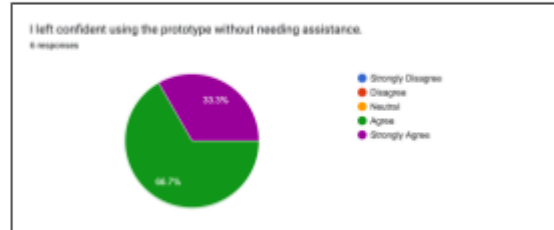
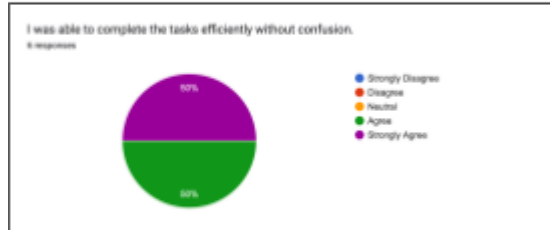
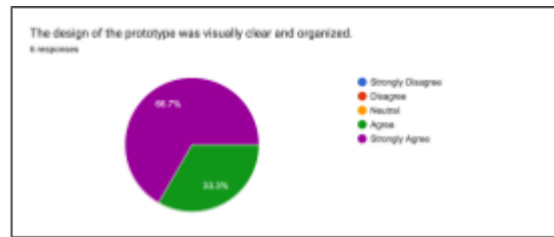
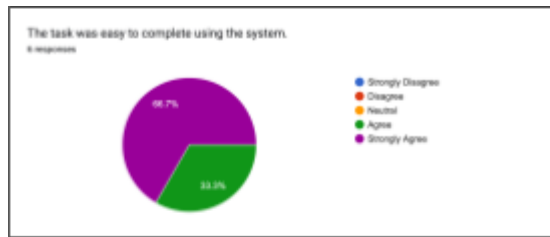
Task 2



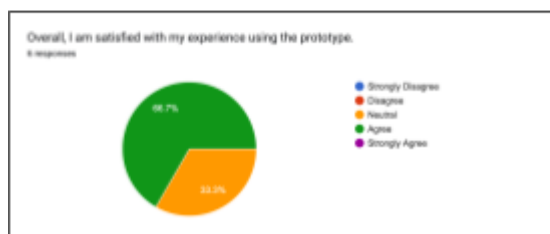
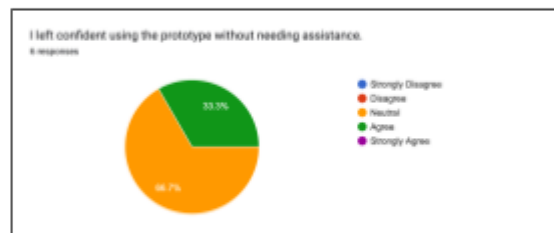
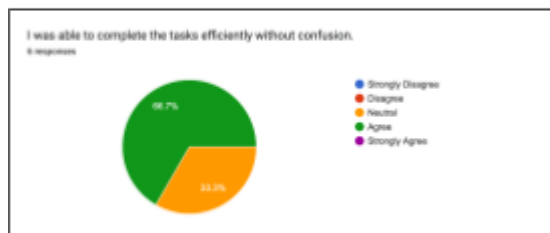
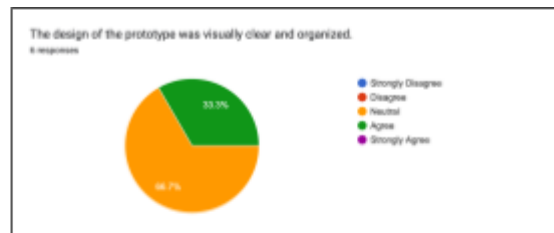
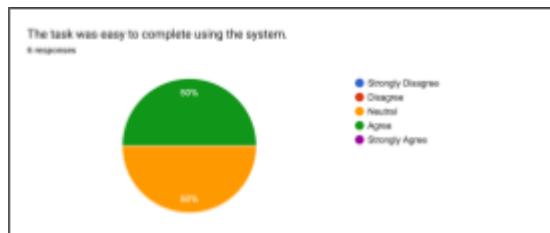
Task 3



Task 4



Task 5



Satisfaction Table:

Task	Question 1 Avg	Question 2 Avg	Question 5 Avg	Overall
1	4.67	4.50	4.50	4.56
2	4.50	4.33	4.50	4.44
3	2.33	2.50	2.17	2.33
4	4.67	4.67	4.83	4.72
5	3.50	3.33	3.67	3.50

12.3.4 Summary of Data

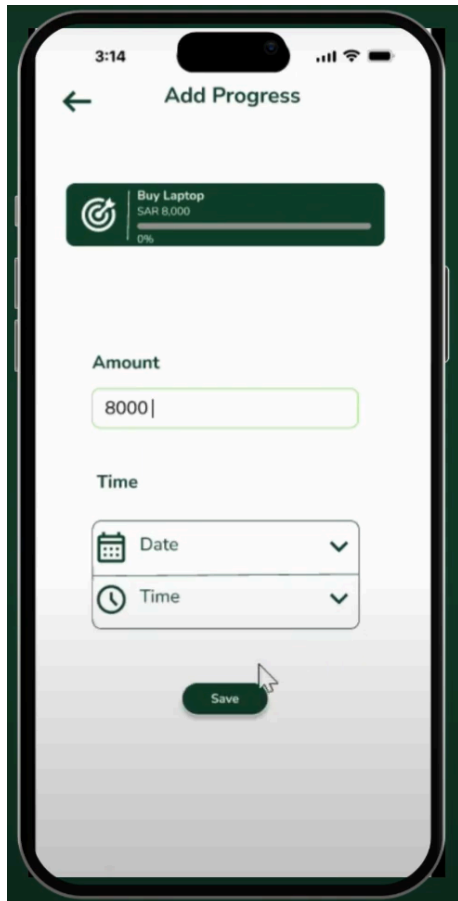
the table below represents a summary of the test data. Low satisfaction rates, high errors and time on tasks are highlighted in red.

Task	Task Completion	Errors	Time on Task	Satisfaction*
1	6	0	297	4.56
2	6	0	270	4.44
3	6	3	185	2.33
4	6	0	321	4.72
5	6	0	159	3.50

12.4 Discussion

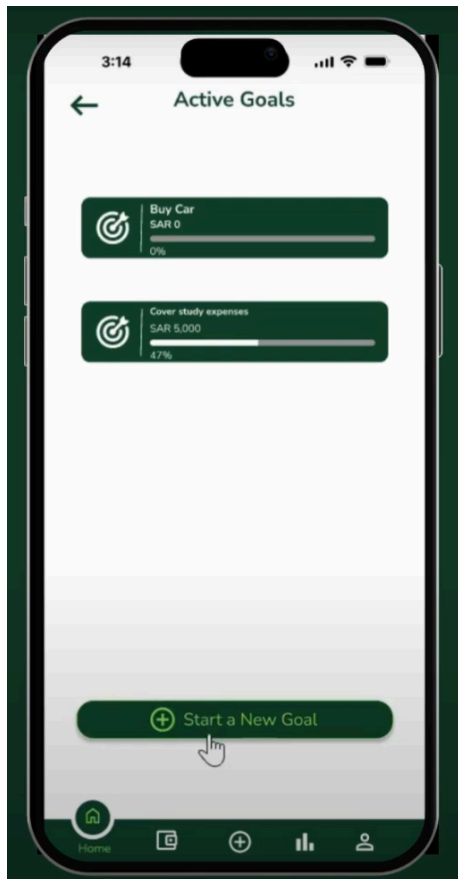
12.4.1 critical problem

Lack of feedback after selecting a date and time: Users are facing confusion after determining the date and the time, as the interface does not give a clear indicator of the user's choice. The users are usually unsure whether a date or time has been selected at all, or if their inputs has been registered correctly, which is a clear problem in the user experience, especially with respect to synthesizability, as users can not asses the system's state based on their actions.



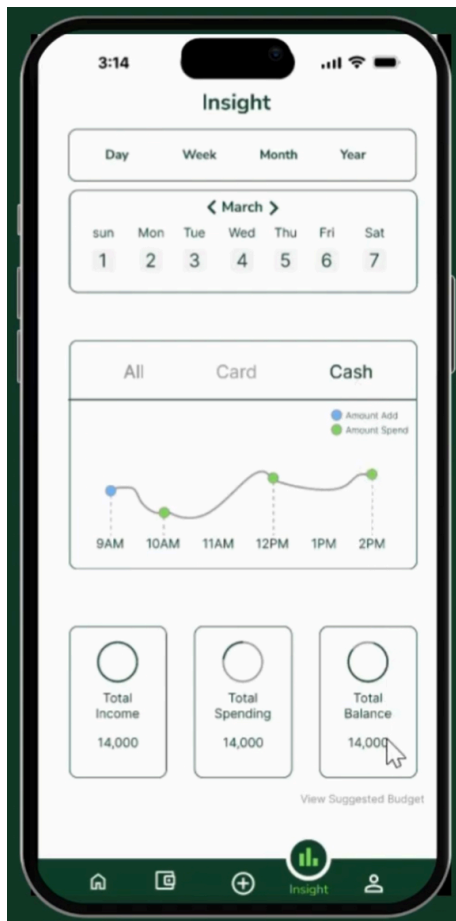
12.4.2 major problem

A major issue in the app is the lack of confirmation when setting a new goal. Users were unsure if their goal was saved, as no message or visual feedback appeared. This caused confusion and repeated actions. Clear confirmation is needed to improve trust and usability.



12.4.3 Minor problem

Task 3, some users had difficulty noticing the “View Suggested Budget” button in the Insight page. The button was placed at the bottom right and had a very light color that made it less visible. We recommend increasing the font size and using a darker color to improve its visibility.



12.5 Recommendations

After completing the usability testing and analyzing participant feedback, several key suggestions emerged to improve the functionality and user experience of the “Mudhakhir” application. One major recommendation was to enhance the visibility of the “View Suggested Budget” button by increasing its size and adjusting its color. Navigation should also be simplified by using clearer labels, intuitive icons, and maintaining consistency across all screens to reduce user confusion. Additionally, participants emphasized the need for real-time feedback after task completion, such as confirmation messages or visual indicators, to increase user confidence. Other improvements include adding error-prevention tools, undo options, and accessibility features like adjustable text size and high-contrast modes for older users. Finally, users recommended adding a comprehensive financial dashboard that visually summarizes income, expenses, savings, and goals to provide a clearer financial overview. These enhancements aim to make Mudhakhir more intuitive, accessible, and helpful for users at all levels.

13. Conclusion

We began our project by conducting user research to understand the users' needs and the challenges they face in their daily lives. This started with eight interviews, which helped us build a strong foundation for identifying our target users' pain points and goals. Following the interviews, we distributed an online questionnaire to reach a wider and more diverse group of participants aged 18 and above. Based on the results of both the questionnaire and selected interviews, we created three user personas that helped us empathize with our future users. We then developed four user stories, which made it easier to envision how the application could assist people in real-life situations.

To move forward with clarity and confidence, we sketched initial interface ideas to support seamless user interaction. We also created two storyboards to visualize how the application would help users in different scenarios. These were followed by a user flow diagram based on the sketches, allowing us to map the application's functionality and structure. We analyzed our design in light of established usability principles to ensure ease of use.

Using our sketches and design plan, we built an interactive prototype that brought our ideas to life. We ensured that it was both functional and goal-oriented. To support the prototype, we also created a video walkthrough and uploaded it to YouTube to showcase the application and explain its features.

With the prototype ready, we planned and conducted a usability study. We selected five key tasks that represent the main functions of our application, wrote a satisfaction questionnaire to evaluate usability, prepared a facilitator script, and created a consent form for ethical participation. Six participants took part in the study. After signing the consent forms, they completed the tasks while thinking aloud, and finally filled out the questionnaire.

Looking ahead, we are committed to continuing the development of this project. Based on the feedback we received during the usability test, we plan to work on enhancing visibility, consistency, and real-time feedback — as well as addressing all user suggestions. Our goal is to apply everything we've learned in this course to keep improving the app and deliver a user-friendly experience.

Appendix A

Interview Questions

What challenges do you face when managing your daily expenses?

What do you do when you exceed your monthly budget?

How can security or privacy issues be resolved when using an expense tracking app?

How do you currently track your expenses?

What features would help you control your spending?

How can an expense management app help you achieve your long-term financial goals?

How do you like to receive expense notifications?

What factors influence your trust in expense management apps?

How do you prefer your financial reports to be presented to you for better clarity?

What is one feature you would like to have in an expense tracking app?

Interview's transcript

Interview 1:

Interviewer: Amal

Interviewee: Reem

I am a university student, balancing my university scholarship. I manage a limited income and am looking for ways to better track and control my expenses.

Interview Answers:

A1: I struggle with tracking small daily expenses like coffee and transport. Unexpected costs, like textbooks, also make budgeting hard.

A2: I cut back on non-essentials like eating out and shopping. If needed, I use savings or borrow from a friend.

A3: The app should use strong encryption, allow biometric login, and avoid storing sensitive bank details.

A4: I mostly check my bank app for my balance. Sometimes, I write expenses in notes, but I don't do it consistently.

A5: Spending limits with alerts, daily expense summaries, and automatic categorization would help me manage my money better.

A6: It can track my spending habits, set savings goals, and send alerts when I exceed my budget, helping me stay financially disciplined.

A7: I prefer push notifications because they're instant and easy to see. Emails are less effective since I check them less often.

A8: I trust apps with strong security, no hidden fees, good reviews, and a simple, ad-free interface.

A9: I prefer clear charts and graphs with a breakdown of spending categories. A simple monthly summary would also be helpful.

A10: An automatic expense tracker that syncs with my bank and categorizes spending without manual input.

Interview 2:

Interviewer: Amal

Interviewee: Rana

I am a beginner employee who recently started working full-time after graduation. I manage a steady income but am still learning how to manage my monthly expenses and savings effectively.

Interview Answers:

A1: I struggle with budgeting for both fixed expenses and small daily purchases. It's hard to balance saving while covering bills, especially with unexpected costs popping up.

A2: I try to cut back on unnecessary purchases, like dining out. If it's really tight, I might borrow from family or reduce non-essential spending for the next month.

A3: The app should have strong security, like encryption and two-factor authentication. I'd also feel safer if my financial data isn't stored long-term.

A4: I use my bank's app to check my balance, but I don't have a proper system to track all my spending.

A5: I would like spending limits, alerts when I'm getting close to them, and easy-to-read spending summaries.

A6: It could help me stick to a budget, track my spending, and set goals for saving. I need something that keeps me on track without complicating things too much.

A7: I prefer getting push notifications, as I check them instantly on my phone.

A8: I trust apps that have clear privacy policies and strong security. A simple design without too many ads makes me feel more comfortable using it.

A9: I prefer simple charts or graphs that show where my money is going. A clear monthly summary would help me see how I'm doing with my budget.

A10: I'd like an app that automatically categorizes my expenses and syncs with my bank account.

Interview 3:

Interviewer: Fajr

Interviewee: Jana

I am an entrepreneur, running my own business and handling both personal and business finances. I focus on growth, managing expenses, and making strategic financial decisions.

Interview Answers:

A1: I carefully review the expenses to see what can be cut or adjusted. If needed, I reallocate funds from other categories or use my business profits to balance it out.

A2: Managing both personal and business expenses at once can get tricky. Unexpected business costs often impact my personal budget, making it hard to balance everything.

A3: The app should use encryption and two-factor authentication. It should not store sensitive financial data and must have regular security updates.

A4: I use a combination of my bank's app and an Excel sheet to track both personal and business expenses. I try to update it weekly, but it can get overwhelming at times.

A5: Having a clear overview of my expenses by category, automatic alerts when I approach my budget limits, and the ability to set savings goals would really help me manage my spending.

A6: An app can help by tracking both personal and business expenses, setting savings targets, and providing insights to improve cash flow. It would keep me on track towards my long-term financial goals.

A7: I prefer receiving push notifications directly on my phone for immediate updates. It helps me stay aware of my spending in real-time.

A8: I trust apps that offer strong security, clear privacy policies, and ease of use. Positive user reviews and no hidden fees are also important factors for me.

A9: I prefer financial reports with clear charts and graphs, showing spending trends and categorized expenses. A simple, monthly summary would make it easy to understand.

A10: I'd love a feature that automatically syncs with my bank accounts and categorizes my expenses, saving me time on manual tracking.

Interview 4:

Interviewer: Fajr

Interviewee: Ahmed

I am a father, managing family expenses while balancing work and personal life. I focus on budgeting for household needs and planning for my children's future.

Interview Answers:

A1: Balancing household needs, kids' expenses, and unexpected costs can be tough. It's hard to stick to a budget with so many things to account for.

A2: I carefully review the expenses to see what can be cut or adjusted. If needed, I reallocate funds from other categories or use my business profits to balance it out.

A3: For family expenses, security is crucial. The app should have encryption for sensitive data, offer two-factor authentication, and ensure that no personal information is stored without permission. Regular updates are also important for maintaining security.

A4: I currently track my expenses using a simple spreadsheet for family spending. I also check our bank statements regularly to keep an eye on larger transactions.

A5: Having a budget-setting feature with alerts when I'm close to exceeding it would be helpful. I'd also appreciate automatic categorization for household, grocery, and utility expenses.

A6: An app can help by tracking both household expenses and savings goals, giving me a clear picture of our finances. It can also send reminders to stay within budget and help us save for long-term goals like education or retirement.

A7: I prefer receiving push notifications for immediate updates, especially when we're close to exceeding our budget or when a large expense is made.

A8: I trust apps that are secure, user-friendly, and offer transparent privacy policies. Positive reviews from other parents and clear features with no hidden fees also play a big role.

A9: I prefer financial reports that show clear breakdowns by category, like groceries, bills, and savings. Visual aids like graphs and a simple monthly summary help me quickly understand where we stand.

A10: I'd like an automatic syncing feature with bank accounts, so all transactions are categorized and tracked without manual input. This would save time and keep everything organized.

Interview 5:

Interviewer: Ibtisam

Interviewee: Nora

I am a senior employee, with many years of experience in my field. I focus on managing my finances efficiently while preparing for retirement and future needs.

Interview Answers:

A1: As an older employee, I find it challenging to keep track of my expenses, especially with so many automatic payments for bills. I also have difficulty adapting to new apps, so I tend to rely more on paper records, which can be time-consuming.

A2: When I exceed my budget, I cut back on discretionary spending like dining out and entertainment. I also review my expenses to see where I can adjust for the next month.

A3: To address security, the app should use strong encryption and offer two-factor authentication. I also prefer apps that don't store sensitive financial data for extra privacy.

A4: I mainly track my expenses using a paper ledger and bank statements. I'm not very comfortable with digital tracking, but I check my account regularly.

A5: Automatic expense categorization and reminders when I'm close to exceeding my budget would help me stay on track. A simple interface would also make it easier to use.

A6: An app could help me set savings goals, track my spending, and alert me when I'm going over budget, which would make it easier to plan for retirement and other long-term needs.

A7: I prefer receiving notifications via email, as I check it more often. Simple, clear alerts about overspending or upcoming bills would be most helpful.

A8: I trust apps that are easy to use, have clear privacy policies, and are well-reviewed. Security features like encryption and two-factor authentication are also important to me.

A9: I prefer simple reports with clear breakdowns of spending categories. Charts or graphs that show trends over time would also help me understand my finances better.

A10: I would like an app that automatically syncs with my bank account and categorizes expenses without requiring manual input. This would save me time and reduce errors.

Interview 6:

Interviewer: Ibtisam

Interviewee: Sara

I am an investor, focused on managing and growing my portfolio. I aim to make informed financial decisions to secure long-term wealth and achieve my financial goals.

Interview Answers:

A1: As an investor, I juggle multiple streams of income, which can make it hard to track every little expense. I often find it challenging to keep my personal and investment expenses separate.

A2: I review my spending to identify areas where I can cut back, like discretionary purchases. If needed, I shift funds from my investment returns or savings.

A3: The app should use strong encryption and offer two-factor authentication. It's also important that the app doesn't store sensitive data long-term and has a clear privacy policy.

A4: I use a combination of spreadsheets and finance apps to track my personal expenses and investments. I also rely on my bank's app for an overview of my spending.

A5: Having spending limits, alerts when nearing my budget, and automatic categorization of expenses would be very helpful.

A6: The app can help by tracking my spending and investments, providing insights into where I can save more, and allowing me to set goals for retirement or large investments. Regular reminders and progress reports would keep me on track.

A7: I prefer receiving push notifications on my phone for immediate updates. Unlike emails, they distract me.

A8: I trust apps that have strong security measures, a clean interface, and no hidden fees. Positive reviews from other investors and clear, transparent policies also play a key role in building my trust.

A9: I prefer reports with clear charts and graphs that show trends over time. A summary with categorized expenses and investment performance would make it easier to understand my financial situation at a glance.

A10: I would like an automatic sync feature that links my investment accounts and bank transactions, categorizing everything automatically to save time and reduce errors.

Interview 7:

Interviewer: Jawaher

Interviewee: Dana

I am a retiree, managing my finances after leaving the workforce. I focus on budgeting for daily expenses while ensuring my savings last for the long term.

Interview Answers:

A1: I find it hard to manage my expenses with a fixed income. Unexpected medical or household costs often throw off my budget.

A2: I reduce spending on non-essential items like entertainment. Sometimes, I adjust my savings for the month.

A3: The app should use strong encryption and require passwords or biometrics. I'd feel safer if it doesn't store my bank details.

A4: I use a simple notebook and check my bank statements monthly. It's not the most efficient, but it works for me.

A5: Automatic categorization of expenses and reminders when I approach my limit would be very helpful for me.

A6: It could help me track my savings and set up monthly goals. I need something to keep me on track for retirement planning.

A7: I prefer simple, clear notifications on my phone. I don't check emails as often, so push notifications work best.

A8: I trust apps that are easy to use, have good reviews, and provide privacy assurances. I also prefer apps that don't have too many ads.

A9: I like straightforward reports with simple charts that break down my spending into categories like food, utilities, and savings.

A10: I'd like an automatic sync with my bank account, so my expenses get updated without me having to manually enter everything.

Interview 8:

Interviewer: Jawaher

Interviewee: Lana

I am a small business owner, managing both personal and business finances. I focus on growing my business while ensuring financial stability and controlling expenses.

Interview Answers:

A1: Balancing personal and business expenses is difficult. I also struggle with cash flow management, especially during slow months.

A2: I review my expenses, cut non-essential costs, and adjust the next month's budget to compensate for the overspending.

A3: The app should use encryption and two-factor authentication, ensuring that sensitive business data is secure and private.

A4: I use a combination of spreadsheets and my bank's app to track both business and personal expenses.

A5: Spending limits, automatic categorization of expenses, and real-time alerts when I'm close to exceeding my budget would help.

A6: It can help track savings and investments, set financial goals, and provide insights into areas where I can cut back to save more.

A7: I prefer receiving push notifications on my phone for immediate updates and email summaries for a more detailed review.

A8: I trust apps with solid security, transparent privacy policies, ease of use, and good customer reviews.

A9: Simple charts and graphs showing spending trends, along with a clear monthly summary, would be ideal for me.

A10: An automatic sync with my bank and credit cards, categorizing transactions without manual entry, would save me time.

Appendix B

Questionnaire Questions[[The Questionnaire](#)]

What is your age?					
18-24	25-34	35-44	45-54	55-64	Above 64
What is your gender?					
Female			Male		
What are your current financial goals?					
Saving	Paying off debt		Investing	Other goals	
How difficult is it for you to stick to your monthly budget?					
Very difficult	Sometimes difficult		Rarely difficult	Easy	
How helpful is a tool that categorizes expenses automatically?					
Very helpful		Somewhat helpful		Not helpful	
Do you know where most of your money is being spent?					
I know exactly		I somewhat know		I don't know	
Do reminders help you track your expenses?					
Yes, always		Sometimes		I don't need them	
How does having a clear savings plan affect you?					
It motivates me		It helps me a little		It doesn't change anything	
What do you think about visual tools for managing money?					
Very useful		Somewhat useful		Not necessary	
Does your busy schedule affect your ability to track expenses?					
Yes, it does		Sometimes		No, it doesn't	

Individual Responses

Ppt	Q1	Q2	Q3	Q4	Q5
1	25–34	Female	Saving	Sometimes difficult	Very helpful
2	25–34	Female	Saving	Sometimes difficult	Very helpful
3	18–24	Male	Paying off debt	Easy	Somewhat helpful
4	35–44	Female	Investing	Rarely difficult	Very helpful
5	55–64	Male	Other goals	Sometimes difficult	Very helpful
6	45–54	Female	Investing	Easy	Somewhat helpful
7	35–44	Male	Saving	Very difficult	Very helpful
8	25–34	Female	Investing	Rarely difficult	Very helpful
9	18–24	Male	Paying off debt	Easy	Somewhat helpful
10	45–54	Female	Saving	Very difficult	Very helpful
11	25–34	Male	Other goals	Sometimes difficult	Not helpful
12	35–44	Female	Investing	Rarely difficult	Very helpful
13	18–24	Male	Paying off debt	Easy	Somewhat helpful
14	18–24	Female	Saving	Sometimes difficult	Very helpful
15	25–34	Male	Paying off debt	Rarely difficult	Somewhat helpful
16	35–44	Female	Investing	Very difficult	Very helpful
17	45–54	Male	Paying off debt	Sometimes difficult	Somewhat helpful
18	55–64	Female	Saving	Easy	Not helpful
19	18–24	Male	Saving	Rarely difficult	Very helpful
20	35–44	Female	Other goals	Very difficult	Somewhat helpful

Ppt	Q6	Q7	Q8	Q9	Q10
1	I somewhat know	Yes, always	It motivates me	Very useful	Sometimes
2	I somewhat know	Sometimes	It motivates me	Very useful	Yes, it does
3	I know exactly	Yes, always	It helps me a little	Somewhat useful	Sometimes
4	I know exactly	I don't need them	It motivates me	Very useful	No, it doesn't
5	I know exactly	Yes, always	It helps me a little	Very useful	No, it doesn't
6	I don't know	Sometimes	It motivates me	Not necessary	Yes, it does
7	I somewhat know	Yes, always	It motivates me	Somewhat useful	Sometimes
8	I know exactly	Sometimes	It helps me a little	Very useful	Yes, it does
9	I know exactly	I don't need them	It motivates me	Somewhat useful	No, it doesn't
10	I somewhat know	Yes, always	It motivates me	Very useful	Sometimes
11	I don't know	Sometimes	It doesn't change anything	Not necessary	Yes, it does
12	I know exactly	I don't need them	It motivates me	Very useful	No, it doesn't
13	I know exactly	Yes, always	It helps me a little	Somewhat useful	Sometimes
14	I somewhat know	Yes, always	It motivates me	Very useful	Yes, it does
15	I know exactly	Sometimes	It helps me a little	Somewhat useful	Sometimes
16	I know exactly	Yes, always	It motivates me	Very useful	Yes, it does
17	I somewhat know	Sometimes	It helps me a little	Somewhat useful	Sometimes
18	I know exactly	I don't need them	It motivates me	Not necessary	No, it doesn't
19	I somewhat know	Sometimes	It motivates me	Very useful	Sometimes
20	I know exactly	Yes, always	It motivates me	Very useful	Yes, it does

Appendix C

Facilitator Script

Hello, and thank you for participating in this usability testing session.

Today, you'll be interacting with a prototype we designed as part of a course project. The goal of this session is to evaluate the usability of the system.

You will be asked to complete five tasks using the system.

Please try to complete each task as you naturally would. If you feel confused at any point, feel free to say it out loud, this helps us understand the user experience better.

I'll be observing and taking notes, but I won't be giving you any help or guidance during the tasks.

Once you finish each task, I'll ask you to fill out a short satisfaction questionnaire.

Before we begin

- There is no right or wrong way to complete the tasks.
- You may stop at any time if you feel uncomfortable.
- We're evaluating the system, not you.

If you're ready, we can begin.

Satisfaction Questionnaire Form

Please check the option that best matches your opinion for each statement below:

Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The task was easy to complete using the system.					
I understood clearly what the task was asking me to do.					
The steps required to complete the task were clear and straightforward.					
I felt confident while performing the task.					
Overall, I am satisfied with how the system helped me do this task.					

Consent Form

Goal:

The goal of this evaluation is to test the usability of a digital prototype I designed, in order to improve its user-friendliness and overall interaction experience.

Procedure:

You will be asked to complete 5 simple tasks using the prototype provided. After each task, you will be given a short questionnaire to share your feedback.

Throughout the session, you are encouraged to think aloud and share any thoughts or confusion you experience.

The full session is expected to take around 10–15 minutes.

You may stop, take a break, or ask any question at any moment.

Recording and Privacy:

There are no risks or consequences related to your participation.

- Your answers will be processed anonymously.
- The goal is to evaluate the system, not your performance.
- All responses will be used for academic and design improvement purposes only.

Do you agree with recording?

- ☐ Yes, video and audio recording
- ☐ Yes, only audio recording
- ☐ No, I prefer not to be recorded

Informed Consent:

By signing this form, you confirm that you have read and understood the details above and agree to take part in this usability testing session.

Name: _____

Date: _____

Signature: _____